



SatQuery AI

RESEARCH
CLOUD



ASSISTANT

+ New Chat

History

Presentation Mode

ANALYSIS

Single Image

Optical + SAR

Bi-temporal

Grounding

Mission & Sensor
Comparison

WORKSPACE

Reports

Benchmarks

Demo Gallery

Research Analytics

SLI Compliance

EVIDENCE-FIRST REMOTE-SENSING INTELLIGENCE

Ask Earth Anything.

Upload one or two Earth observations, choose a workflow, and ask a question. SatQuery returns an evidence-grounded result with limitations and provenance intact.

SatQuery Agent Ready

System health
6/6 specialists ready

01 · WORKFLOW

Choose how SatQuery should read the observations

Optical + SAR

Combine spectral and structural evidence.



Single Image

SELECT

Describe scenes, answer questions, or localize objects.

One optical, multispectral, or SAR observation
Is water visible?



Optical + SAR

ACTIVE

Combine spectral and structural evidence.

One optical image and one SAR observation
Where do both modalities agree?



Bi-temporal

SELECT

Detect and explain changes between observations.

Earlier and later observations with dates
What changed between these dates?

CURRENT QUERY Where do both modalities agree?



ACTIVE WORKFLOW

Optical + SAR Joint Analysis

Combine spectral information from optical imagery with structural SAR evidence.

ACTIVE WORKFLOW

MODELS

DEVELOPER

SYSTEM

SATELLITE IMAGERY

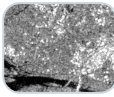
Uploaded Satellite Images

Hide imagery

SATELLITE IMAGERY

Optical observation

RGB or multispectral view

 **sa...**
0.05 MB · Input verified

Compatible Input

No Georeference

Sar Preview

PNG · 256 × 256 · Sar Preview

Scientific metadata

Display preview only — not a scientific product.

• Display preview only — not a scientific product.

SATELLITE IMAGERY

SAR observation

Radar observation · VV/VH or supported preview

 **op...**
0.14 MB · Input verified

Compatible Input

No Georeference

Optical Rgb

PNG · 256 × 256 · Optical Rgb

Scientific metadata

Display preview only — not a scientific product.

• Display preview only — not a scientific product.

Detected workflow **Optical + SAR Pair**

Review detected modalities

ANALYTICAL QUERY

Ask SatQuery

Questions are answered from published specialist evidence.

Question

Where do both modalities agree?

31 / 2,000 · ⌘/Ctrl + Enter

TRY ASKING

Where do both modalities agree?

Is water visible in both observations?

What does SAR reveal that optical does not?

Compare built-up areas.

Where is structural evidence strongest?

Describe the scene using both modalities.

Reuse compatible analysis

Run Analysis

EXECUTIVE SUMMARY

Where do both modalities agree?

✓ COMPLETED

Low · 398 ms

PDF

JSON

ZIP

Compare

Judge View

Research View

DIRECT ANSWER

COMPLETED

Display Preview — Qualitative Analysis Only

**Pixel-level optical–SAR
answering is unavailable
because exact co-
registration was not
established. No joint
percentage was fabricated.**

Query: Where do both modalities agree?

► What determines confidence?

WORKFLOW

**Optical +
SAR**

STATUS

COMPLETED

RUNTIME

398 ms

CONFIDENCE

Low

PRIMARY SPECIALIST

**Cross Modal
Optical Sar
Analyzer**

SUPPORTING SPECIALISTS

**None
published**

PDF, JSON, and Full ZIP actions stay visible in the sticky result header and are generated from the authoritative backend record. Compare opens stored-result comparison.



REMOTE-SENSING ADAPTATION

Remote-sensing adapted

Ready

SatQuery Vision Encoder v1

OpenCLIP ViT-L/14 · Adapted on BigEarthNet.txt

DEVICE

mps

RUNTIME

352 ms

CAPTION CONSISTENCY

Not requested

VQA CONSISTENCY

Not requested

SIMILARITY-BASED SCENE PRIORS · NOT PROBABILITIES

urban or built-up area 0.278

residential area 0.259

industrial area 0.255

harbor 0.236

airport 0.194

Optical-to-generated-RGB semantic consistency: Skipped Raw Sar. Raw SAR is not encoded. A validated generated RGB-like product is required for this supporting comparison.

► Limitations and scientific disclosure

SOURCES

Evidence Sources

Only sources published by the completed workflow are listed.



Cross-modal analyzer

Agent-selected workflow source
Cross Modal Optical Sar Analyzer

USED



SatQuery Vision Encoder

Remote-sensing scene evidence
SatQuery Vision Encoder v1 · SatQuery Vision Encoder v1 provides scene-level embedding evidence. It does not produce pixel-level segmentation, object grounding, calibrated probabilities, or ground truth.

USED



Visual question answering

Primary answer source
Deterministic Optical-SAR Evidence Statistics

USED



Evidence participation is derived from the response execution trace and published result fields.

02 Evidence

Source imagery and the most relevant backend-produced evidence first.

Evidence viewer

Second authoritative observation supplied to the workflow.

Layer

Opacity

Split

+

-

↺

↻

Evidence opacity 0%

SAR source



Optical source

Primary Evidence

SAR source



Primary Evidence

03 Key Statistics

Measured values remain distinct from calibrated probabilities.

Dimensions

256 × 256

Bands

1

Georeferenced

No

Runtime

398 ms

Evidence products

2

04 Why This Answer?

Published evidence, routing rationale, confidence, and limitations.

Backend-published rationale

-  The selected Optical + SAR workflow requests source-aware joint or complementary evidence.
-  The protected cross-modal engine did not produce joint statistics.

Data

- No georeference
- Pixel alignment is assumed from the supplied pairing; geospatial co-registration cannot be independently verified.
- Display preview only — not a scientific product.

Model

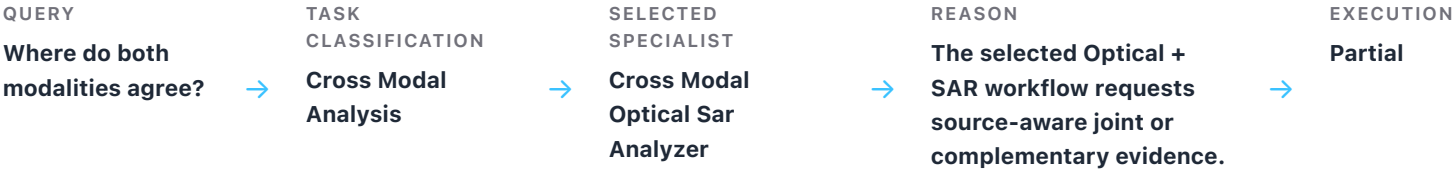
- This deterministic baseline is not a trained or calibrated land-cover classifier.
- Model output is not ground truth

Interpretation

- No pixel-level fusion, evidence masks, or joint statistics were computed.
- Water-like and structural-likelihood evidence can be confused by shadow, terrain, speckle, and strong scatterers.

- Optical and SAR percentages are deterministic candidate evidence, not calibrated semantic probabilities.

- Dark smooth terrain, cloud shadow, terrain shadow, and radar shadow can resemble water-like evidence.
- Bright terrain, speckle, vegetation structure, and isolated strong scatterers can resemble built-up or structural evidence.



05

Execution Timeline

Every observable backend stage, grouped for review without hiding the underlying record.

Input

5 stages · 0 ms

✓

Upload

0 ms · Success

✓

Upload

0 ms · Success

✓

Metadata

0 ms · Success

✓

Metadata

0 ms · Success

✓

Validation

0 ms · Success

Validation

7 stages · 0 ms

✓

Pair Compatibility Check

0 ms · Success

7

Optical Preparation

0 ms · Skipped

8

Sar Preparation

0 ms · Skipped

✓

SVE Eligibility Check

0 ms · Success



SatQuery Agent Ready

6/6 specialists ready



Dark interface

Multimodal remote sensing intelligence



SVE Image Preparation

0 ms · Success



SVE Routing Support

0 ms · Success

22

SVE Eligibility Check

0 ms · Skipped

Model Inference

5 stages · 352 ms

Evidence Fusion

4 stages · 0 ms

Response

1 stage · 0 ms